**Supplementary information**

# Ectopic cambia in wisteria vines are associated with the expression of conserved KNOX genes

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**Tables not included in this file are provided as separate files:**

- Supplementary Table 2
- Supplementary Table 4
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- Supplementary Table 11

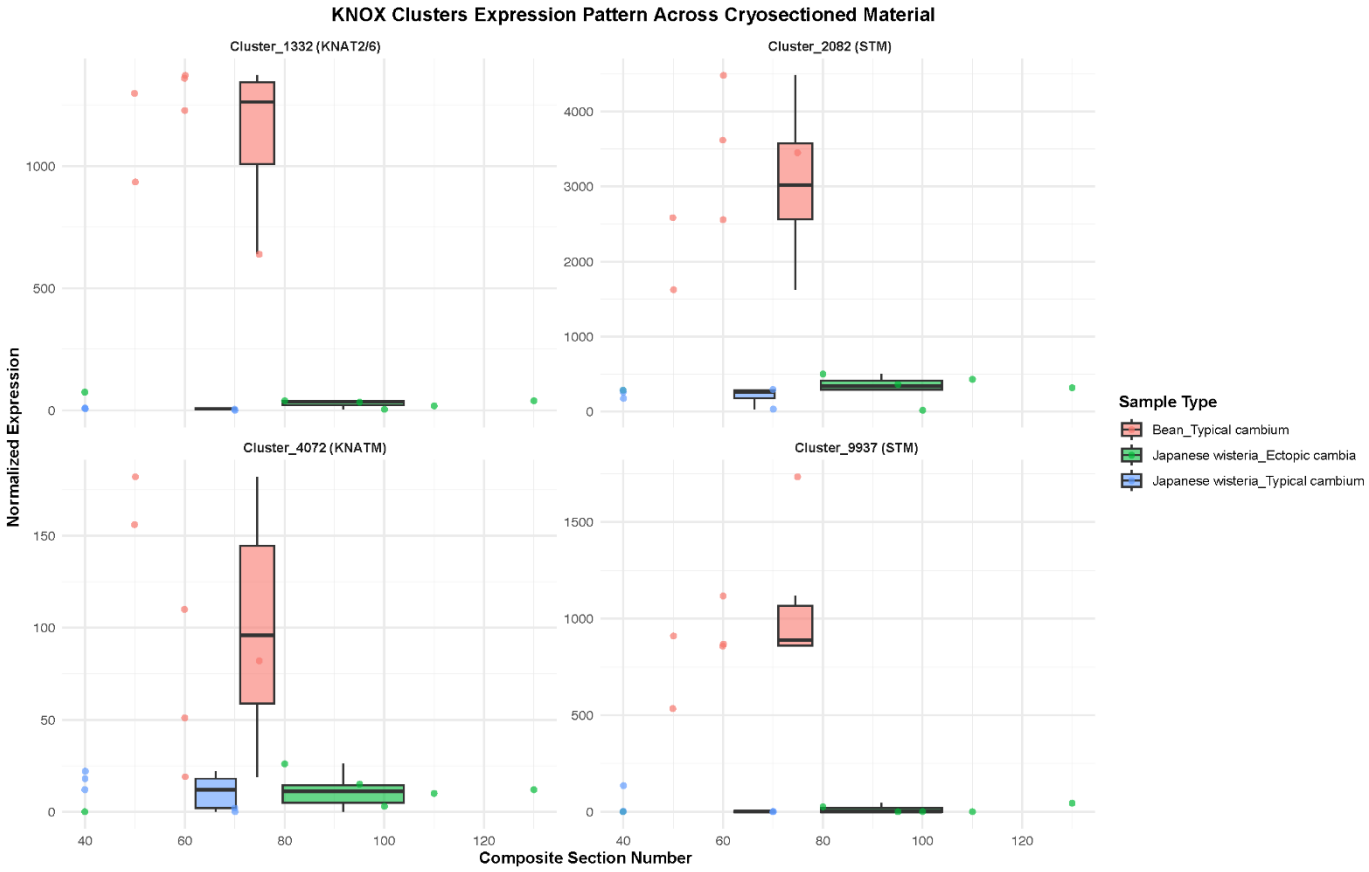
**Supplementary Table 1.** The transcriptome assembly statistics (i.e., BUSCO, Mapping Rate, and N50) for the final transcriptome assemblies for *Phaseolus vulgaris* (common bean) and *Wisteria floribunda* (Japanese wisteria).

| Species | BUSCO | Mapping Rate | N50 |
|---|---|---|---|
| Common bean | C:81.8%[S:77.1%,D:4.7%], F:3.9%, M:14.3% | 82.41% | 1562 |
| Japanese Wisteria | C:88.4%[S:79%,D:9.4%], F:1.6%, M:10% | 84.99% | 1624 |

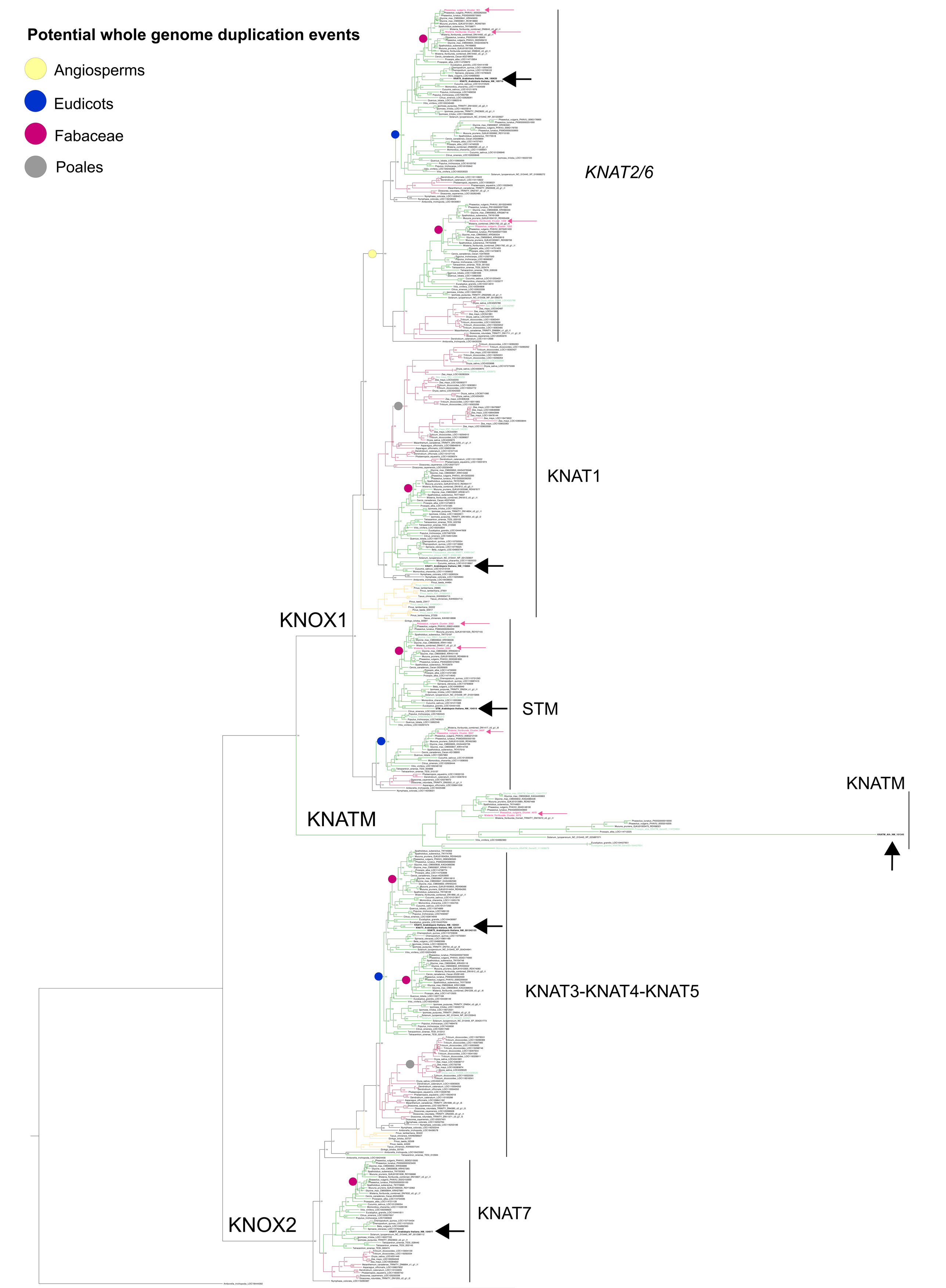
**Supplementary Table 3.** The significantly enriched gene ontology (GO) terms (< 0.05 FDR) for differentially expressed transcript clusters in the "Ectopic Cambia v. Typical Cambia of Japanese wisteria and Common bean" contrast. All p-values were adjusted for multiple testing burden using a Benjamini-Hochberg procedure based on a two-sided test.

| Term | Gene Ontology | Direction | p-value |
|---|---|---|---|
| mRNA processing | Biological Process | ▲ | 0.0002 |
| mRNA Splicing, via Spliceosome | Biological Process | ▲ | 0.0007 |
| RNA Splicing | Biological Process | ▲ | 0.04 |
| UV Protection | Biological Process | ▲ | 0.04 |
| Spliceosomal Complex | Cellular Component | ▲ | 0.002 |
| Nucleolus | Cellular Component | ▲ | 0.004 |
| Nucleus | Cellular Component | ▲ | 0.006 |
| RNA binding | Molecular Function | ▲ | 0.000005 |
| mRNA binding | Molecular Function | ▲ | 0.003 |
| nucleic acid binding | Molecular Function | ▲ | 0.04 |
| DNA-directed 5'-3' RNA polymerase activity | Molecular Function | ▲ | 0.05 |

Footnote: Each term includes its ontology and corrected p-value rounded to one significant figure. "Ectopic Cambia v. Typical Cambia of Japanese wisteria and Common bean" is the only contrast with enriched GO terms using the differentially expressed transcript clusters. The filled arrows indicate if the terms were significantly overrepresented (▲).




**Supplementary Figure 1. Relative expression of KNOX genes across composite cryosection numbers for three cambia samples.** Cambium and its derivatives (xylem and phloem) were cryosectioned and pooled together (Fig. 2), so the section number reflects a composite radial axis rather than a defined tissue position. Interpretation should focus on relative trends within sample types rather than absolute tissue-specific localization or the total number of sections. Note also that RNA-seq data were subsampled to equal read counts per individual; the results are proportional to each other, and the number of sections does not alter the number of reads. The analysis was conducted with six biological replicates for the typical cambium of common bean and ectopic cambia (EC) of Japanese wisteria, and five replicates for the typical cambium of Japanese wisteria.

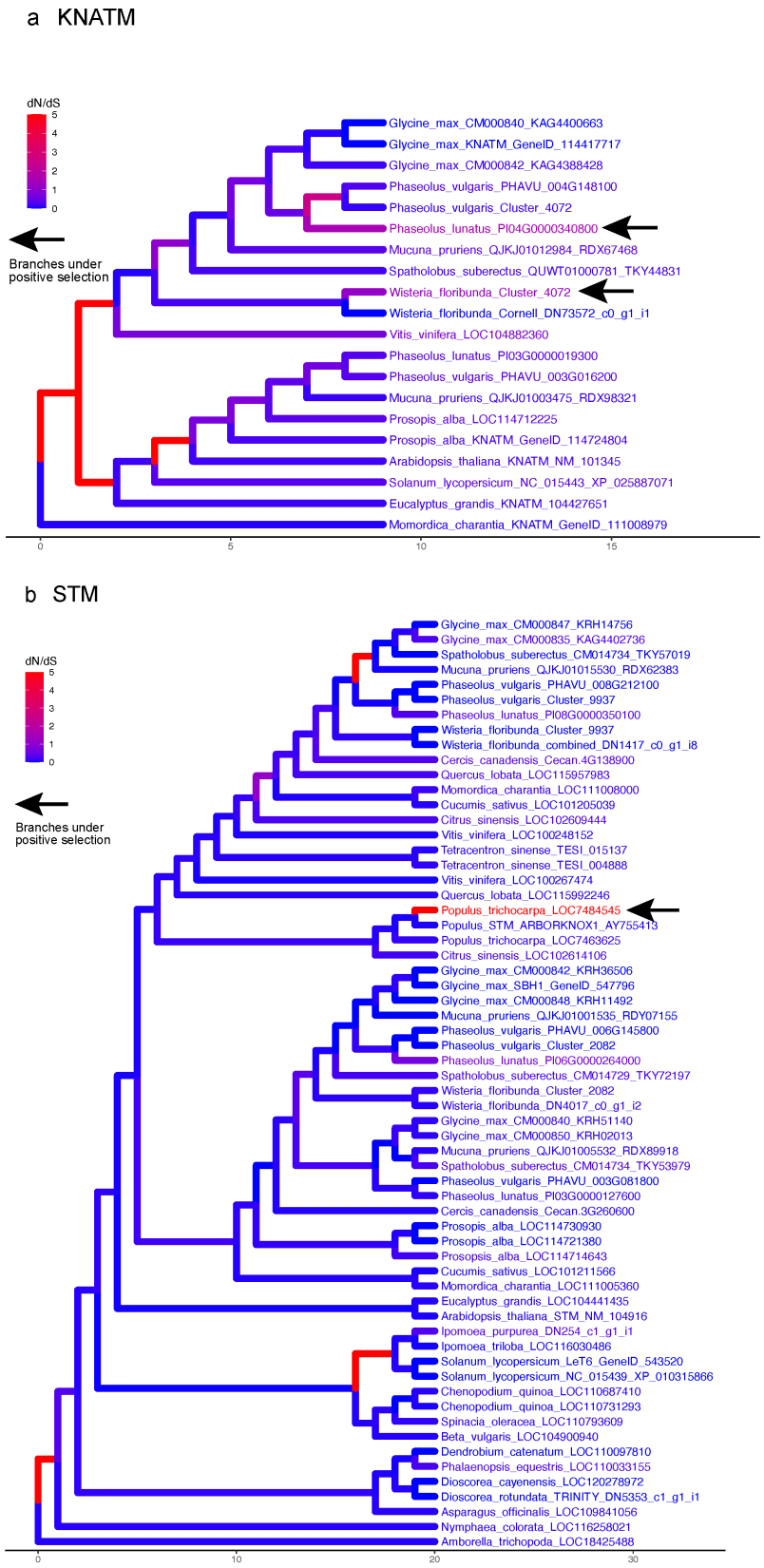



**Supplementary Figure 2**. Schematic phylogeny of the KNOX gene family indicating the three classes with respective genes. Numbers indicate bootstrap value. Branches of major clades are color-coded as follows: eudicots (green), monocots (dark red), *Amborella* +*Nymphaea* (black), and gymnosperms (yellow). Black arrows indicate the position of *Arabidopsis thaliana* references, and names of additional references included in the phylogeny are shown in light green (see also Supplementary Table 4 and Supplementary Table 5). The sequences differentially expressed in our study are shown in pink (indicated with arrows). Circles indicate potential whole-genome duplications, as shown in Fig. 3 and Fig. 4.

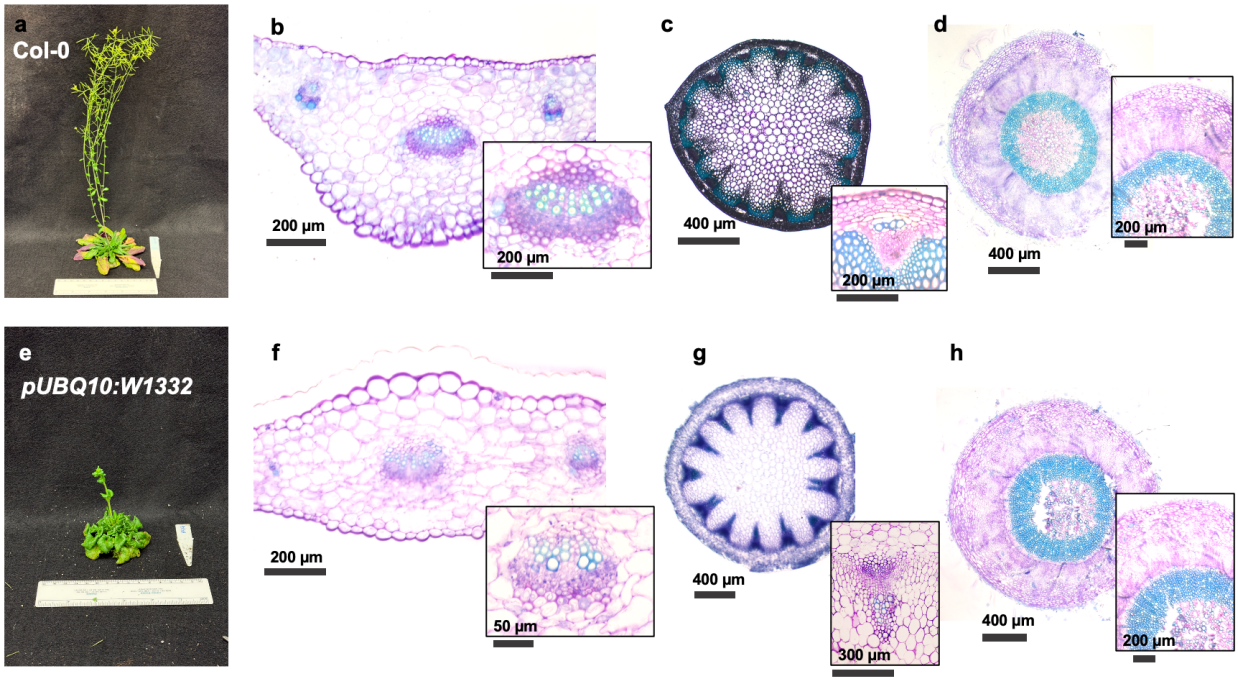
**This clade includes likely other genes (not KNOX), belonging to contigs from the following species:**
Cycas panzhihuaensis
Gnetum montanum
Liriodendron chinense
Magnolia biondii
Asparagus officinale,
Cordyline australis
Yucca gloriosa
Zingiber officinale

**KNOX2**

**KNOX1 (including KNATM)**




**Supplementary Figure 4**. **Rooted tree fitted for branch models for the three genes investigated under selection analysis.** The arrows indicate the branches under positive selection, *i.e.*, omega values greater than 1 (ω >1). **a** KNATM. **b** STM. Arrows indicate branches under positive selection.




**Supplementary Figure 5**. **Anatomical analysis of wild and transformed *Arabidopsis thaliana* plants for a heterologous expression essay.** All plants were sampled at ~90 days postgermination. **a-d** Wildtype. **e-h** Transformants carrying *pUBQ:W1332.* **a,e** Plant collected for anatomical study. Ruler = 15 cm. **b,f** leaf (midrib), **c,g** inflorescence stem, **d,h** root. All anatomical images (b-g) feature an overall view of the organ and a close-up of the vascular system, confirming the absence of any vascular abnormalities.

**Supplementary Methods 1**

**Building Supplementary Tables 4**

To construct **Supplementary Table 4**, we utilized two gene sets as queries across referenced papers. The first set comprises nine KNOX genes identified in *Arabidopsis thaliana*, also used in phylogenetic analysis (see Methods). The second set includes 20 genes enriched in single-cell transcriptome analysis of vascular meristems in *Populus*[1]. Gene names (acronyms in column C of Supplementary Table 4) were obtained by querying the NCBI database for gene names (column D) and loci (column E). Column F information was sourced from the Gene Ontology database (https://amigo.geneontology.org/) using *A. thaliana* locus IDs (column E), emphasizing ontology descriptions linked to ectopic cambia (EC) development. Original annotations are in Supplementary Table 1 and referenced studies.

Selected genes (by locus number or acronyms in column C) were cross-referenced with differential gene expression databases from our study and seven others focusing on cambium in *A. thaliana* and *Populus*. These papers fall into two groups: (1) studies without mutant lines, such as the wild *Populus* cambium transcriptome[2] and cambium regeneration after girdling[3], detailed further in **Supplementary Table 5** due to relevance to EC. (2) Studies with mutant lines including *KNAT1* (=*ARK2*) in *Populus* and *A. thaliana* mutants reported with EC[4–8]. For studies involving multiple mutants, the gene expression of each line is displayed in separate columns. Studies without differential gene expression data were not included[9].

**Supplementary Methods 2**

**TransPi Pipeline**

All de novo transcriptomes were constructed with the TransPi pipeline[10]. Briefly, assemblies were generated for each dataset using rnaSPAdes version 3.14[11], SOAPdenovo-Trans version 1.03[12], TRANS-ABYSS version 2.0.1[13], Velvet version 1.2.12/Oasis version 0.2.09[14,15] for kmer values 25, 35, 45, 55, and 65 along with Trinity version 2.9.1 [16] before being merged and reduced with the EvidentialGene pipeline version 2019.05.14[17]. The EvidentialGene pipeline reduces the complexity of the transcriptomes by collapsing duplicates, clustering protein sequences, and performing similarity searches between the transcripts with a combination of CD-HIT[18] and BLAST version 2.2.31[19]. Quality assessment was done using TransRate version 1.0.3 (Smith-Unna et al. 2016) for assembly statistics, Hisat2 version 2.2.1[20] to assess mapping efficiency and Benchmarking Universal Single-Copy Orthologs (BUSCO) version 5.5.0cv1 to evaluate the completeness of expected universal single copy genes with the fables database[21]. For annotation, TransDecoder version 5.5.0[22] was employed to denote open reading frames (ORFs) and complete homology searches of the ORFs against proteins via BLAST to preserve ORFS with potential functional significance). HMMER version 3.3[23] was used to complete protein domain searches against the PFAM database. Trinotate version 3.2.0[24] was used for functional annotation and DIAMOND version 0.9.30[25] was used for the similarity searches between the transcripts and the UniProt database for fables (UniProt Taxon ID 72025[26]) The final annotation report from Trinotate included the evolutionary genealogy of genes: Non-supervised Orthologous Groups (eggNOG), Kyoto Encyclopedia of Genes and Genomes (KEGG), and Gene Ontology (GO) as well as protein alignments.

**Supplementary Methods 3**

**Limitations of the one-to-one orthologs approach**

It is important to note that a limitation of working with transcriptomes instead of genomes is that our clusters may not accurately represent genes, and we have been careful to refer to these as transcript clusters to reflect this uncertainty. Genes may have orthologs across species, but the corset clustering might split a real gene into two clusters, affecting our ability to identify one-to-one orthologs. It is also important to note that not all genes are likely to be annotated in the transcriptomes. Because RNA-seq investigates relative

gene expression, reads were subset to a standard 20 million reads before quantification, which could reduce the counts of rare genes. Additionally, when filtering for lowly expressed genes by biological treatment, a threshold (i.e., 10 counts) is set for the number of reads that need to be aligned to a given transcript cluster in at least two replicates for it to be retained for downstream analysis. Therefore, if at least two samples in the species do not have at least 10 reads aligned to that transcript cluster, it is discarded. Generally, these are considered genes with extremely low expression and do not contain enough information to be statistically valid. Therefore, three main scenarios may lead to conserved genes not being represented in the final analysis: (1) not assigning one-to-one orthologs due to gene splitting across clusters, (2) unsuccessful protein annotations, and (3) no differential expression in the current analysis, which may be affected by filtering.

**Supplementary Methods 4**

**Summary of steps to generate KNOX phylogeny**

(Files are available on https://github.com/anthonysnead/Fabaceae-Ectopic_Cambia_Transcriptomics).

| Step | Description | Notes | # of sequences |
|---|---|---|---|
| Step_1 | Original output from BLAST pipepline | Includes additional sequences from GenBank (see Table S6), and the transcript clusters from our differential expression analysis. | 5571 |
| Step_2 | Deleted duplicated sequences using sRNAtoolbox | 4660 sequences were removed. | 911 |
| Step_3 | Deleted BEL1 genes, duplicated isoforms, and small sequences. | Small sequences (<450 bp) were deleted from all taxa, except the Fabaceae. In total, 255 sequences were removed. | 656 |
| Step_4 | Deleted an additional 39 duplicated contigs. | | 617 |
| Step_5 | Running MACSE | We aligned the previous fasta file using MAFFT and ran MACSE and then trimAL + raxML to generate a tree. | 617 |
| Step_6 | Deleted remaining BEL1 contigs. | BELL contigs were deleted based on their phylogenetic position. In total, 92 sequences were deleted. One BELL sequence was maintained to root the tree. | 525 |
| Step_7 | Detected contigs that poorly align and that likely does not belong to KNOX genes. | A tree is shown to illustrate this clade (see **Supplementary Figure 3**. | 525 |
| Step_8 | Deleted the “non-KNOX” group mentioned above and ran a new MACSE alignment. | | 497 |
| Step_9 | Deleted 31 gappy sequences (with ~1/3 of missing data). | See the list of deleted sequences in **Supplementary Figure 10**, which include also the sequences listed under Step_3 and Step_4. | 466 |

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